



Original Article

Dislocation-oxide interaction in Y_2O_3 embedded Fe: A molecular dynamics simulation studyM. Mustafa Azeem^a, Qingyu Wang^{a,*}, Zhongyu Li^a, Yue Zhang^b^a College of Nuclear Science and Technology, Fundamental Science on Nuclear Safety and Simulation Technology Laboratory, Harbin Engineering University, Harbin, 150001, China^b Nuclear and Radiation Safety Center, MEE, Beijing, 100082, China

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ABSTRACT

Oxide dispersed strengthened (ODS) steel is an important candidate for Gen-IV reactors. Oxide embedded in Fe can help to trap irradiation defects and enhances the strength of steel. It was observed in this study that the size of oxide has a profound impact on the depinning mechanism. For smaller sizes, the oxide acts as a void; thus, letting the dislocation bypass without any shear. On the other hand, oxides larger than 2 nm generate new dislocation segments around themselves. The depinning is similar to that of Orowan mechanism and the strengthening effect is likely to be greater for larger oxides. It was found that higher shear deformation rates produce more fine-tuned stress-strain curve. Both molecular dynamics (MD) simulations and BKS (Bacon-Knocks-Scattergood) model display similar characteristics whereby establishing an inverse relation between the depinning stress and the obstacle distance. It was found that $(110)_{\text{oxide}} \parallel (111)_{\text{Fe}}$ (oriented oxide) also had similar characteristics as that of $(100)_{\text{oxide}} \parallel (111)_{\text{Fe}}$ but resulted in an increased depinning stress thereby providing greater resistance to dislocation bypass. Our simulation results concluded that critical depinning stress depends significantly on the size and orientation of the oxide.

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1. Introduction

The materials in advanced fission and fusion reactors entail higher-level strength in order to endure stringent irradiation criterion. Irradiation effects in structural materials are initiated by incident particles that result in production of point defects and dislocations. The atomic scale modeling provides a comprehensive mechanism to study properties of these materials and their defect interaction with obstacles. Therefore, a thorough understanding of defects' evolution can help in the selection of better radiation resistant materials, suitable for extended service conditions.

Fe-based ODS alloys are considered as one of the most promising candidate materials for advanced fission and fusion reactors due to their higher thermal stability, high temperature creep and corrosion resistance. It has been reported that radiation can affect the size of oxide along-with creation of amorphization [1–3]. Researchers concluded that oxide particles having ranges up to 3 nm act as coherent solute-enriched clusters while larger sizes act as

near-stoichiometric complex particles [4]. As we know that dislocation obstacle interactions have profound effect on its mechanical properties. Therefore, mechanical properties and radiation resistance response of ODS steel significantly relies upon the size of oxide and achieving a higher understanding of dislocation nucleation in the radiation environment is crucial for determining the strength of any material [5,6]. The phenomenon of precipitate hardening enhance structural strengthening, but it increase embrittlement [7]. For the same reason, nature and mechanism of dislocation hindrance through presence of nano-oxides needs to be thoroughly explained [8,9].

Stress-driven dynamics of dislocation have been a topic of interest during the past decade [10,11]. Two kinds of interaction mechanisms have been reported in literature when investigating temperature effects [12]. One kind is that of the Orowan mechanism which is repulsive in nature while the second type exhibits attraction commonly at extreme temperatures in accordance with Srolovitz mechanism [12,13]. Climbing has also been observed when dislocation encounters a hard or non-shearable obstacle.

Terentyev et al. [14] mentioned that nature of the obstacle determines the interaction mechanism. They have found that the

* Corresponding author.

E-mail address: wangqingyu@hrbeu.edu.cn (Q. Wang).